

# UV-LED Maskers 4" & 6"

## UV-KUB Z

### UV-LED Flood Exposure & Masking System



**“Benefits of UV-LED source:  
low power consumption, very long lifetime”**

Min. < 2 $\mu$ m feature size

Up to 4" or 6" exposure area

365nm UV-LED source, +/- 5nm

10.000 hours of LEDs lifetime

2 exposure modes: continuous or flashing

Compatible with all UV photoresists sensitive to 365nm

### *Benefits*

*No warm-up nor cold down required*

*Touch-screen operated*

*Equipment fully closed*

*No need to operate in cleanroom*

# Handy & compact equipment, **UV-KUB Z** is equipped with a 365nm monochromatic & collimated UV Light source based on LED technology.

## High quality collimation

**Divergence angle below 2°** offers the possibility to work in masking mode on thick resist layers without altering the rendering of edges of the patterns.

High-quality collimation allows to reach resolution down to the micrometre scale **without needing vacuum contact mode**.

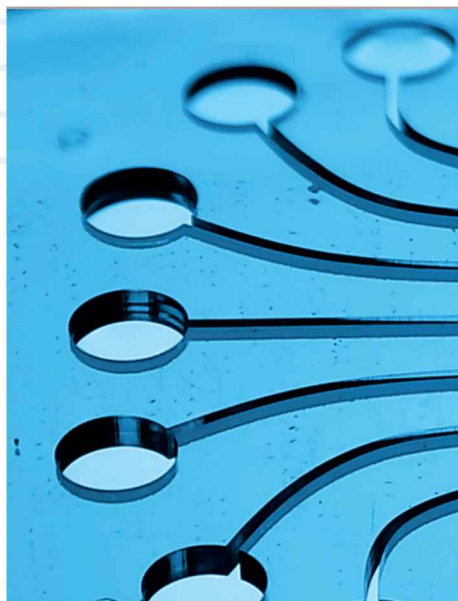
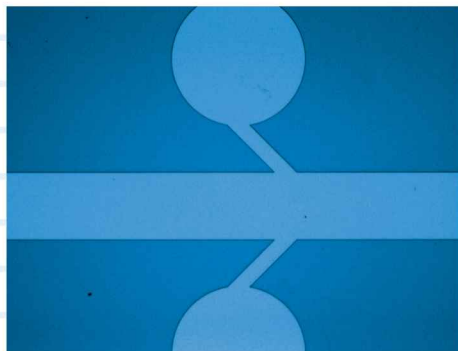
## Led technology

Perfectly **monochromatic** exposure light source.

**Cold UV source**, prevents undesirable thermal effects.

Operate in **continuous or flashing mode**.

**Benefits of UV-LED source:** low power consumption, very long lifetime.



## PERFORMANCES

|                                        |                                             |
|----------------------------------------|---------------------------------------------|
| Minimum feature size                   | < 2µm                                       |
| Divergence half-angle                  | < 2°                                        |
| Number of programmable cycles          | 10                                          |
| Exposure cycles (continuous or cyclic) | From 1s to 1h59min                          |
| Contact modes                          | Hard (physical) or Soft (proximity) contact |

## EXPOSURE AREA

|                               |                                                        |        |
|-------------------------------|--------------------------------------------------------|--------|
| Maximum exposure area         | 4"                                                     | 6"     |
| Maximum substrate size        | Ø 4"                                                   | Ø 6"   |
| Maximum mask size             | Ø 5" *                                                 | Ø 7" * |
| Substrate warm-up during exp. | < 1°C                                                  |        |
| Compatible film photomask     | Chrome and flexible film photomasks                    |        |
| Compatible photoresists       | Compatible with all UV photoresists sensitive to 365nm |        |

\*Optional adaptaters available on request

## UV-LED SOURCE

|                       |                      |
|-----------------------|----------------------|
| Available wavelengths | 365nm +/- 5nm        |
| Power density (RMS)   | Min. 40mW/cm² +/- 5% |
| Exposure homogeneity  | +/- 5%               |
| LEDs Lifetime         | > 10 000 hours       |

## USAGE FEATURES

|                            |                                                      |
|----------------------------|------------------------------------------------------|
| Color touchscreen          | 5.7"                                                 |
| Exposure modes             | Continuous or Flashing                               |
| Included equipment         | UV exposure area fully closed                        |
| Cleanroom type environment | Not required, due to its dust-proof protective cover |

## GENERAL SPECIFICATIONS

|                       |                                      |
|-----------------------|--------------------------------------|
| Size (L x W x Ht)     | 260 x 260 x 260mm                    |
| Weight                | 8,2kg / 18lbs                        |
| Power supply          | 100-110V / 220-240V - 50Hz           |
| Power consumption     | 180W                                 |
| Operating temperature | From 15° to 30°C / Stability +/- 1°C |

